The key is that the PDP can always show where each signal originated and that it arrived unaltered. Provenance, not blind trust, is what gives the authorization system confidence in the data it uses for policy evaluation.

12.5.3 Limiting signal scope

Preserving confidentiality means limiting how far signals travel and ensuring each system sees only the data it needs. Currently, ACME's various integrations each decide which attributes to pull and store for their own purposes. The Customer Collaboration platform requests HR and Finance data relevant to document access; the engineering deployment tool queries PagerDuty and device posture status. ACME lacks corporate guidance on which attributes each PDP should access. Each integration makes its own decision.

That autonomy works in the short term but introduces inconsistency and risk. Some systems may gather more data than necessary, others less, and some may retain context longer than intended. These differences eventually led ACME to develop a unified enterprise PIP and normalization layer, discussed later in this chapter, to standardize which signals are collected, how they're transmitted, and who can access them.

As ACME found out, signal delivery isn't just a technical detail—it's a crucial part of the trust model. The authorization system must be able to verify that every piece of data used in a decision comes from a reliable source and hasn't been changed. If policy is the logic behind authorization, then secure transmission is its nervous system: it delivers the context that makes authorization decisions possible.